PART I
Clinical Decision Making

OUTLINE

1. Evaluation of the Urologic Patient: History and Physical Examination
2. Evaluation of the Urologic Patient: Testing and Imaging
3. Urinary Tract Imaging: Basic Principles of CT, MRI, and Plain Film Imaging
4. Urinary Tract Imaging: Basic Principles of Urologic Ultrasonography
5. Urinary Tract Imaging: Basic Principles of Nuclear Medicine
6. Assessment of Urologic and Surgical Outcomes
7. Ethics and Informed Consent

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Evaluation of the Urologic Patient

History and Physical Examination

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Keywords

digital rectal examination; history; physical examination

Questions

1. Pain associated with a stone in the ureter is the result of:
    a. obstruction of urine flow with distention of the renal capsule.
    b. irritation of the ureteral mucosa by the stone.
    c. excessive ureteral peristalsis in response to the obstructing stone.
    d. irritation of the intramural ureter.
    e. urinary extravasation from a ruptured calyceal fornix.
2. The most common cause of gross hematuria in a patient older than 50 years is:
    a. renal calculi.
    b. infection.
    c. bladder cancer.
    d. benign prostatic hyperplasia.
    e. trauma.
3. The most common cause of pain associated with gross hematuria is:

a. simultaneous passage of a kidney stone.
b. ureteral obstruction due to blood clots.
c. urinary tract malignancy.
d. prostatic inflammation.
e. prostatic enlargement.

4. All of the following are typical lower urinary tract symptoms associated with benign prostatic hyperplasia EXCEPT:
   a. urgency.
   b. frequency.
   c. nocturia.
   d. dysuria.
   e. weak urinary stream.
5. The most likely cause of continuous incontinence (loss of urine at all times and in all positions) is:
   a. enterovesical fistula.
   b. noncompliant bladder.
   c. detrusor hyperreflexia.
   d. vesicovaginal fistula.
   e. sphincteric incompetence.
6. All of the following are potential causes of anejaculation EXCEPT:
   a. sympathetic denervation.
   b. pharmacologic agents.
   c. bladder neck and prostatic surgery.
   d. androgen deficiency.
   e. cerebrovascular accidents.
7. What percentage of patients with multiple sclerosis will present with urinary symptoms as the first manifestation of the disease?
   a. 1%
   b. 5%
   c. 10%
   d. 15%
   e. 20%
8. What important information is gained from pelvic bimanual examination that cannot be obtained from radiologic evaluation?
   a. Presence of bladder mass
   b. Invasion of bladder cancer into perivesical fat

c. Presence of bladder calculi
d. Presence of associated pathologic lesion in female adnexal structures
e. Mobility/fixation of pelvic organs

9. With which of the following diseases is priapism most commonly associated?
a. Peyronie disease
b. Sickle cell anemia
c. Parkinson disease
d. Organic depression
e. Leukemia

10. What is the most common cause of cloudy urine?
a. Bacterial cystitis
b. Urine overgrowth with yeast
c. Phosphaturia
d. Alkaline urine
e. Significant proteinuria

11. Conditions that decrease urine specific gravity include all of the following EXCEPT:
a. increased fluid intake.
b. use of diuretics.
c. decreased renal concentrating ability.
d. dehydration.
e. diabetes insipidus.

12. Urine osmolality usually varies between:
a. 10 and 200 mOsm/L.
b. 50 and 500 mOsm/L.
c. 50 and 1200 mOsm/L.
d. 100 and 1000 mOsm/L.
e. 100 and 1500 mOsm/L.

13. Elevated ascorbic acid levels in the urine may lead to false-negative results on a urine dipstick test for:
a. glucose.
b. hemoglobin.
c. myoglobin.
d. red blood cells.

e. leukocytes.

14. Hematuria is distinguished from hemoglobinuria or myoglobinuria by:
    a. dipstick testing.
    b. the simultaneous presence of significant leukocytes.
    c. microscopic presence of erythrocytes.
    d. examination of serum.
    e. evaluation of hematocrit.
15. The presence of one positive dipstick reading for hematuria is associated with significant urologic pathologic findings on subsequent testing in what percentage of patients?
    a. 2%
    b. 10%
    c. 25%
    d. 50%
    e. 75%
16. The most common cause of glomerular hematuria is:
    a. transitional cell carcinoma.
    b. nephritic syndrome.
    c. Berger disease (immunoglobulin A nephropathy).
    d. poststreptococcal glomerulonephritis.
    e. Goodpasture syndrome.
17. The most common cause of proteinuria is:
    a. Fanconi syndrome.
    b. excessive glomerular permeability due to primary glomerular disease.
    c. failure of adequate tubular reabsorption.
    d. overflow proteinuria due to increased plasma concentration of immunoglobulins.
    e. diabetes.
18. Transient proteinuria may be due to all of the following EXCEPT:
    a. exercise.
    b. fever.
    c. emotional stress.
    d. congestive heart failure (CHF).
    e. ureteroscopy.

19. Glucose will be detected in the urine when the serum level is above:
    a. 75 mg/dL.
    b. 100 mg/dL.
    c. 150 mg/dL.
    d. 180 mg/dL.
    e. 225 mg/dL.
20. The specificity of dipstick nitrite testing for bacteriuria is:
    a. 20%.
    b. 40%.
    c. 60%.
    d. 80%.
    e. >90%.
21. All of the following are microscopic features of squamous epithelial cells EXCEPT:
    a. large size.
    b. small central nucleus.
    c. irregular cytoplasm.
    d. presence in clumps.
    e. fine granularity in the cytoplasm.
22. The number of bacteria per high-power microscopic field that corresponds to colony counts of 100,000/mL is:
    a. 1.
    b. 3.
    c. 5.
    d. 10.
    e. 20.
23. Pain in the flaccid penis is usually due to:
    a. Peyronie disease.
    b. bladder or urethral inflammation.
    c. priapism.
    d. calculi impacted in the distal ureter.
    e. hydrocele.
24. Chronic scrotal pain is most often due to:
    a. testicular torsion.
    b. trauma.

c. cryptorchidism.
d. hydrocele.
e. orchitis.

25. Terminal hematuria (at the end of the urinary stream) is usually due to:
    a. bladder neck or prostatic inflammation.
    b. bladder cancer.
    c. kidney stones.
    d. bladder calculi.
    e. urethral stricture disease.
26. Enuresis is present in what percentage of children at age 5 years?
    a. 5%
    b. 15%
    c. 25%
    d. 50%
    e. 75%
27. All of the following in the medical history suggest that erectile dysfunction is more likely due to organic rather than psychogenic causes EXCEPT:
    a. sudden onset.
    b. peripheral vascular disease.
    c. absence of nocturnal erections.
    d. diabetes mellitus.
    e. inability to achieve adequate erections in a variety of circumstances.
28. All of the following should be routinely performed in men with hematospermia EXCEPT:
    a. cystoscopy.
    b. digital rectal examination.
    c. serum prostate-specific antigen (PSA) level.
    d. genital examination.
    e. urinalysis.
29. Pneumaturia may be due to all of the following EXCEPT:
    a. diverticulitis.
    b. colon cancer.
    c. recent urinary tract instrumentation.

d. inflammatory bowel disease.
e. ectopic ureter.

30. Which of the following disorders may commonly lead to irritative voiding symptoms?
    a. Parkinson disease
    b. Renal cell carcinoma
    c. Bladder diverticula
    d. Prostate cancer
    e. Testicular torsion

Answers

1. **a. Obstruction of urine flow with distention of the renal capsule.** Pain is usually caused by acute distention of the renal capsule, usually from inflammation or obstruction.
2. **c. Bladder cancer.** The most common cause of gross hematuria in a patient older than age 50 is bladder cancer.
3. **b. Ureteral obstruction due to blood clots.** Pain in association with hematuria usually results from upper urinary tract hematuria with obstruction of the ureters with clots.
4. **d. Dysuria.** Dysuria is painful urination that is usually caused by inflammation.
5. **d. Vesicovaginal fistula.** Continuous incontinence is most commonly due to a urinary tract fistula that bypasses the urethral sphincter or an ectopic ureter.
6. **e. Cerebrovascular accidents.** Anejaculation may result from several causes: (1) androgen deficiency, (2) sympathetic denervation, (3) pharmacologic agents, and (4) bladder neck and prostatic surgery.
7. **b. 5%.** In fact, 5% of patients with previously undiagnosed multiple sclerosis present with urinary symptoms as the first manifestation of the disease.
8. **e. Mobility/fixation of pelvic organs.** In addition to defining areas of induration, the bimanual examination allows the examiner to assess the mobility of the bladder; such information cannot be

obtained by radiologic techniques such as computed tomography (CT) and magnetic resonance imaging (MRI), which convey static images.

9. **b. Sickle cell anemia. Priapism occurs most commonly in patients with sickle cell disease but can also occur in those with advanced malignancy, coagulation disorders, and pulmonary disease, as well as in many patients without an obvious cause.**
10. **c. Phosphaturia.** Cloudy urine is most commonly caused by phosphates in the urine.
11. **d. Dehydration.** Conditions that decrease specific gravity include (1) increased fluid intake, (2) diuretics, (3) decreased renal concentrating ability, and (4) diabetes insipidus.
12. **c. 50 and 1200 mOsm/L.** Osmolality is a measure of the amount of solutes dissolved in the urine and usually varies between 50 and 1200 mOsm/L.
13. a. **Glucose.** False-negative results for glucose and bilirubin may be seen in the presence of elevated ascorbic acid concentrations in the urine.
14. **c. Microscopic presence of erythrocytes.** Hematuria can be distinguished from hemoglobinuria and myoglobinuria by microscopic examination of the centrifuged urine; the presence of a large number of erythrocytes establishes the diagnosis of hematuria.
15. **c. 25%.** Investigators at the University of Wisconsin found that 26% of adults who had at least one positive dipstick reading for hematuria were subsequently found to have significant urologic pathologic findings.
16. **c. Berger disease (immunoglobulin A nephropathy).** IgA nephropathy, or Berger disease, is the most common cause of glomerular hematuria, accounting for about 30% of cases.
17. b. **Excessive glomerular permeability due to primary glomerular disease.** Glomerular proteinuria is the most common type of proteinuria and results from increased glomerular capillary permeability to protein, especially albumin. Glomerular proteinuria occurs in any of the primary glomerular diseases such as IgA

nephropathy or in glomerulopathy associated with systemic illness such as diabetes mellitus.

18. **e. Ureteroscopy.** Transient proteinuria occurs commonly, especially in the pediatric population, and usually resolves spontaneously within a few days. It may result from fever, exercise, or emotional stress. In older patients, transient proteinuria may be due to CHF.
19. **d. 180 mg/dL.** This so-called renal threshold corresponds to a serum glucose level of about 180 mg/dL; above this level, glucose will be detected in the urine.
20. **e. >90%.** The specificity of the nitrite dipstick test for detecting bacteriuria is greater than 90%.
21. **d. Presence in clumps.** Squamous epithelial cells are large, have a central small nucleus about the size of an erythrocyte, and have an irregular cytoplasm with fine granularity.
22. **c. 5.** Therefore five bacteria per high-power field in a spun specimen reflect colony counts of about 100,000/mL.
23. **b. Bladder or urethral inflammation.** Pain in the flaccid penis is usually secondary to inflammation in the bladder or urethra, with referred pain that is experienced maximally at the urethral meatus.
24. **d. Hydrocele.** Chronic scrotal pain is usually related to noninflammatory conditions such as a hydrocele or varicocele, and the pain is usually characterized as a dull, heavy sensation that does not radiate.
25. **a. Bladder neck or prostatic inflammation.** Terminal hematuria occurs at the end of micturition and is usually secondary to inflammation in the area of the bladder neck or prostatic urethra.
26. **b. 15%.** Enuresis refers to urinary incontinence that occurs during sleep. It occurs normally in children as old as 3 years but persists in about 15% of children at age 5 and about 1% of children at age 15.
27. **a. Sudden onset.** A careful history will often determine whether the problem is primarily psychogenic or organic. In men with psychogenic impotence, the condition frequently develops rather quickly, secondary to a precipitating event such as marital stress or change or loss of a sexual partner.

28. **a. Cystoscopy.** A genital and rectal examination should be done to exclude the presence of tuberculosis, a PSA assessment and digital rectal examination should be done to exclude prostatic carcinoma, and a urinary cytologic assessment should be done to exclude the possibility of transitional cell carcinoma of the prostate.
29. **e. Ectopic ureter.** Pneumaturia is the passage of gas in the urine. In patients who have not recently had urinary tract instrumentation or a urethral catheter placed, this is almost always due to a fistula between the intestine and bladder. Common causes include diverticulitis, carcinoma of the sigmoid colon, and regional enteritis (Crohn disease).
30. **a. Parkinson disease.** The second important example of nonspecific lower urinary tract symptoms that may occur secondary to a variety of neurologic conditions is irritative symptoms resulting from neurologic disease such as cerebrovascular accident, diabetes mellitus, or Parkinson disease.

Chapter Review

1. IPSS score: 0 to 7 mild symptoms, 8 to 19 moderate symptoms, 20 to 35 severe symptoms.
2. Renal pain radiates from the flank anteriorly to the respective lower quadrant and may be referred to the testis, labium, or medial aspect of the thigh. The pain is colicky (fluctuates). It may be associated with gastrointestinal symptoms due to reflex stimulation of the celiac ganglion.
3. Patients with slowly progressive urinary obstruction with bladder distention often have no pain, despite residual volumes in excess of a liter.
4. Pain of prostatic origin is poorly localized.
5. Scrotal pain may be primary or referred. Pain referred to the testicle originates in the retroperitoneum, ureter, or kidney.
6. Hematuria, particularly in adults, should be regarded as a symptom of malignancy until proven otherwise.

7. Adults normally arise no more than twice a night to void. Urine production increases at night (recumbent position) in older patients and those with cardiac disease, particularly CHF.
8. Postvoid dribbling: Urine escapes into the bulbar urethra and then leaks at the end of micturition. This may be alleviated by perineal pressure following voiding.
9. Those who present with microscopic hematuria and irritative voiding symptoms should be suspected of having carcinoma in situ of the bladder until proven otherwise.
10. Continuous incontinence is most commonly due to ectopic ureter, urinary tract fistula, or totally incompetent sphincter.
11. Hematospermia almost always resolves spontaneously and is rarely associated with any significant urologic pathology.
12. When urinary obstruction is associated with fever and chills, it should be regarded as a urologic emergency.
13. It is always worthwhile to obtain the previous operative report in patients who are to be operated on.
14. If the patient is uncircumcised, the foreskin must be retracted for inspection of the glans.
15. The testes are normally 6 cm in length and 4 cm in width.
16. If one obtains a stool guaiac test (hemoccult) as a screen for colon cancer, two subsequent stool specimens must be obtained for an adequate test. If the hemoccult is positive, the patient should be on a red meat–free diet for 3 days before collection of three specimens.
17. A male urologist should always perform a female pelvic examination with a female nurse in attendance.
18. The bulbocavernosus reflex tests the integrity of this spinal cord reflex involving S2 to S4.
19. A positive dipstick for blood in the urine indicates hematuria, hemoglobinuria, or myoglobinuria. Hematuria is distinguished from hemoglobinuria and myoglobinuria by microscopic examination of the centrifuged urine and identification of red blood cells (more than three red blood cells per high-power field is abnormal).
20. Hematuria of nephrologic origin is frequently associated with proteinuria and dysmorphic erythrocytes.

21. Anticoagulation at normal therapeutic levels does not predispose patients to hematuria.
22. The most accurate method to diagnose urinary tract infection is by microscopic examination of the urine and identifying pyuria and bacteria. This is confirmed by urine culture.
23. The chief complaint is the focus of the visit and is the reason the patient seeks consultation. It should be the lead sentence in the history and physical (H&P).
24. A family history should always include questions about renal and prostate cancer, renal cysts, and stone disease.
25. Priapism occurs most commonly in patients with sickle cell disease but can also occur in those with advanced malignancy, coagulation disorders, or pulmonary disease, as well as in many patients without an obvious cause.
26. On urine dipstick, false-negative results for glucose and bilirubin may be seen in the presence of elevated ascorbic acid concentrations in the urine.
27. Glomerular proteinuria is the most common type of proteinuria and results from increased glomerular capillary permeability to protein, especially albumin. Glomerular proteinuria occurs in any of the primary glomerular diseases such as IgA nephropathy or in glomerulopathy associated with systemic illness such as diabetes mellitus.
28. Five bacteria per high-power field in a spun specimen reflect colony counts of about 100,000/mL.
29. An important example of nonspecific lower urinary tract symptoms that may occur secondary to a variety of neurologic conditions is irritative symptoms resulting from neurologic disease such as cerebrovascular accident, diabetes mellitus, and Parkinson disease.
30. The renal threshold for glucose corresponds to a serum glucose level of about 180 mg/dL; above this level, glucose will be detected in the urine.

2

Evaluation of the Urologic Patient

Testing and Imaging

Erik P. Castle, Christopher E. Wolter, and Michael Woods

keywords

dipstick; evaluation; hematuria; Urinalysis; uroflowmetry

Questions

1. What is the most common cause of cloudy urine?
    a. Bacterial cystitis
    b. Urine overgrowth with yeast
    c. Phosphaturia
    d. Alkaline urine
    e. Significant proteinuria
2. Conditions that decrease urine specific gravity include all of the following EXCEPT:
    a. increased fluid intake.
    b. use of diuretics.
    c. decreased renal concentrating ability.
    d. dehydration.
    e. diabetes insipidus.
3. Urine osmolality usually varies between:
    a. 10 and 200 mOsm/L.

b. 50 and 500 mOsm/L.
c. 50 and 1200 mOsm/L.
d. 100 and 1000 mOsm/L.
e. 100 and 1500 mOsm/L.

4. Elevated ascorbic acid levels in the urine may lead to false-negative results on a urine dipstick test for:
   a. glucose.
   b. hemoglobin.
   c. myoglobin.
   d. red blood cells.
   e. leukocytes.
5. Hematuria is distinguished from hemoglobinuria or myoglobinuria by:
   a. dipstick testing.
   b. the simultaneous presence of significant leukocytes.
   c. microscopic presence of erythrocytes.
   d. examination of serum.
   e. evaluation of hematocrit.
6. The presence of one positive dipstick reading for hematuria is associated with significant urologic pathologic findings on subsequent testing in what percentage of patients?
   a. 2%
   b. 10%
   c. 25%
   d. 50%
   e. 75%
7. The most common cause of glomerular hematuria is:
   a. transitional cell carcinoma.
   b. nephritic syndrome.
   c. Berger disease (immunoglobulin A nephropathy).
   d. poststreptococcal glomerulonephritis.
   e. Goodpasture syndrome.
8. The most common cause of proteinuria is:
   a. Fanconi syndrome.
   b. excessive glomerular permeability due to primary glomerular disease.

c. failure of adequate tubular reabsorption.
d. overflow proteinuria due to increased plasma concentration of immunoglobulins.
e. diabetes.

9. Transient proteinuria may be due to all of the following EXCEPT:
a. exercise.
b. fever.
c. emotional stress.
d. congestive heart failure.
e. ureteroscopy.

10. Glucose will be detected in the urine when the serum level is above:
a. 75 mg/dL.
b. 100 mg/dL.
c. 150 mg/dL.
d. 180 mg/dL.
e. 225 mg/dL.

11. The specificity of dipstick nitrite testing for bacteriuria is:
a. 20%.
b. 40%.
c. 60%.
d. 80%.
e. >90%.

12. All of the following are microscopic features of squamous epithelial cells EXCEPT:
a. large size.
b. small central nucleus.
c. irregular cytoplasm.
d. presence in clumps.
e. fine granularity in the cytoplasm.

13. The number of bacteria per high-power microscopic field that corresponds to colony counts of 100,000/mL is:
a. 1.
b. 3.
c. 5.
d. 10.

e. 20.

14. All of following is true of uroflowmetry EXCEPT:
    a. Qmax >20 mL/s is not consistent with obstruction.
    b. Qmax, mean flow rate, and voided volume are parameters obtained from this study.
    c. 80 mL voided volume is adequate for uroflowmetry.
    d. the study can be performed in sitting and standing positions.
    e. uroflowmetry cannot diagnose the location of obstruction.
15. The following should be given to uncomplicated patients undergoing simple flexible diagnostic cystourethroscopy:
    a. single-dose oral antibiotic following procedure
    b. 3 days of oral antibiotics following procedure
    c. 3 days of oral antibiotics starting the day prior to procedure
    d. nothing
    e. single intramuscular injection of ceftriaxone following procedure
16. What is the most appropriate initial workup for asymptomatic microscopic hematuria (AMH)?
    a. Flexible cystoscopy, urinary cytology, CT urogram, and UroVysion FISH
    b. Flexible cystoscopy and CT urogram
    c. CT urogram and NMP22
    d. Flexible cystoscopy, urinary cytology, and CT urogram
    e. Flexible cystoscopy, renal ultrasound, and urinary cytology

Answers

1. **c. Phosphaturia.** Cloudy urine is most commonly caused by phosphates in the urine.
2. **d. Dehydration.** Conditions that decrease specific gravity include (1) increased fluid intake, (2) diuretics, (3) decreased renal concentrating ability, and (4) diabetes insipidus.
3. **c. 50 and 1200 mOsm/L.** Osmolality is a measure of the amount of solutes dissolved in the urine and usually varies between 50 and 1200 mOsm/L.

4. **a. Glucose.** False-negative results for glucose and bilirubin may be seen in the presence of elevated ascorbic acid concentrations in the urine.
5. **c. Microscopic presence of erythrocytes.** Hematuria can be distinguished from hemoglobinuria and myoglobinuria by microscopic examination of the centrifuged urine; the presence of a large number of erythrocytes establishes the diagnosis of hematuria.
6. **c. 25%.** Investigators at the University of Wisconsin found that 26% of adults who had at least one positive dipstick reading for hematuria were subsequently found to have significant urologic pathologic findings.
7. **c. Berger disease (immunoglobulin A nephropathy).** IgA nephropathy, or Berger disease, is the most common cause of glomerular hematuria, accounting for about 30% of cases.
8. **b. Excessive glomerular permeability due to primary glomerular disease.** Glomerular proteinuria is the most common type of proteinuria and results from increased glomerular capillary permeability to protein, especially albumin. Glomerular proteinuria occurs in any of the primary glomerular diseases such as IgA nephropathy or in glomerulopathy associated with systemic illness such as diabetes mellitus.
9. **e. Ureteroscopy.** Transient proteinuria occurs commonly, especially in the pediatric population, and usually resolves spontaneously within a few days. It may result from fever, exercise, or emotional stress. In older patients, transient proteinuria may be due to congestive heart failure.
10. **d. 180 mg/dL.** This so-called renal threshold corresponds to a serum glucose level of about 180 mg/dL; above this level, glucose will be detected in the urine.
11. **e. >90%.** The specificity of the nitrite dipstick test for detecting bacteriuria is greater than 90%.
12. **d. Presence in clumps.** Squamous epithelial cells are large, have a central small nucleus about the size of an erythrocyte, and have an irregular cytoplasm with fine granularity.
13. **c. 5. Therefore 5 bacteria per high-power field in a spun specimen reflect colony counts of about 100,000/mL.**

14. **c. 80 mL voided volume is adequate for uroflowmetry.** The minimum voided volume that is accepted as a requirement for considering an adequate assessment is at least 100 mL.
15. **d. Nothing.** For patients undergoing simple diagnostic flexible cystoscopy no antibiotic prophylaxis is recommended unless there are extenuating risk factors for infection or recent orthopedic implantation of artificial joints. Refer to American Urological Association (AUA) recommendations on antibiotic prophylaxis for urological procedures.
16. **b. Flexible cystoscopy and CT urogram.** For the initial work-up of AMH, routine urine cytology is not necessary. Cytology is generally utilized in patients with a history of bladder cancer undergoing surveillance or the index of suspicion of a high-grade lesion is present.

Chapter Review

1. A catheterized urine specimen should be obtained in the female patient with a history of recurrent urinary tract infections or suspected contaminated specimen.
2. Hematuria should be stratified into glomerular, nonglomerular, medical, and surgical causes.
3. A dipstick alone is inadequate for the diagnosis of microscopic hematuria.
4. Asymptomatic microscopic hematuria is defined as three or greater RBC/HPF on a properly collected urinary specimen in the absence of an obvious benign cause.
5. Urine dipstick positive for only leukocyte esterase or nitrites but not both should be confirmed with microscopic analysis and urine culture.
6. Serum creatinine and glomerular filtration rate should be ordered when renal obstruction of nephrologic disease is suspected.
7. Prostate-specific antigen is a very sensitive test for prostate conditions such as BPH and prostatitis and correlates most often with prostate volume.

8. Urine cytology is not recommended during the initial evaluation and screening for asymptomatic microscopic hematuria.
9. Urine cytology is very specific for high-grade urothelial carcinoma.
10. Uroflowmetry and assessment of postvoid residual should be ordered when lower urinary tract obstruction is suspected.
11. Urodynamic studies provide information on disorders of storage and voiding.
12. Routine use of antimicrobial prophylaxis is not recommended for office cystourethroscopy, urodynamics, or cystography in the patient without risk factors.
13. Renal ultrasonography can provide basic screening information on the presence of hydronephrosis and medical renal disease but is not an adequate stand-alone study for the workup of hematuria.
14. CT without contrast of the abdomen and pelvis is the gold standard for detecting urinary stones.
15. A KUB is a useful and easy test for the follow-up of existing non-emergent radio opaque urinary stones.

3

Urinary Tract Imaging

Basic Principles of CT, MRI, and Plain Film Imaging

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Key Words

contrast media; conventional radiography; intravenous urography; plain abdominal radiograph; radiation protection

Questions

1. The measure of the potential adverse health effects of ionizing radiation in sieverts (Sv) is known as:
    a. radiation exposure.
    b. absorbed dose.
    c. equivalent dose.
    d. effective dose.
    e. relative radiation levels.
2. The relative radiation level associated with abdominal computed tomography (CT) without and with contrast is:
    a. none.
    b. minimal, less than 0.1 mSv.
    c. low, 0.1 to 1.0 mSv.
    d. moderate, 1 to 10 mSv.

e. high, 10 to 100 mSv.

3. Bladder filling may precipitate autonomic dysreflexia in patients with a spinal cord injury above:
   a. S2.
   b. L4.
   c. T10.
   d. T12.
   e. T6.
4. Radiation exposure diminishes as the square of the distance from the radiation source. An exposure of 9 mSv at 1 foot from the source would be how much at 3 feet from the source?
   a. 0.09 mSv
   b. 1 mSv
   c. 3 mSv
   d. 9 mSv
   e. 27 mSv
5. Type 2 diabetics on oral metformin biguanide hyperglycemic therapy are at risk for biguanide lactic acidosis after exposure to intravascular radiologic contrast media if they:
   a. discontinue metformin 48 hours before the study.
   b. have severe renal insufficiency and take metformin the day of the study.
   c. are given a saline injection while taking metformin.
   d. have normal kidney function and fail to stop metformin 48 hours before the study.
   e. decrease metformin dose and increase other antihyperglycemic agents on the day of the study.
6. All of the following are true EXCEPT:
   a. Patients with a history of asthma are at greater risk of having an adverse reaction to contrast media.
   b. Severe allergic reactions are not dose dependent.
   c. Hyperosmolar contrast media are more likely to cause contrast reactions than are iso-osmolar agents.
   d. The mechanism of action associated with severe idiosyncratic anaphylactoid (IA) reactions is an immunoglobulin E (IgE) antibody reaction to the contrast media.

e. Severe cardiac disease is a risk factor for an adverse reaction to contrast media.

7. After rapidly assessing airway, breathing, and circulation, the medical treatment of choice for a severe, life-threatening adverse drug reaction following exposure to contrast media is:
   a. subcutaneous injection of epinephrine 0.5 mg of 1:10,000 epinephrine.
   b. intravenous injection of 100 mg of methylprednisone.
   c. 0.01 mg/kg of epinephrine (1:10,000 concentration), given intramuscularly in the lateral thigh.
   d. intravenous diphenhydramine, 50 mg.
   e. 0.01 mg/kg of epinephrine (1:1000 concentration), given intramuscularly in the lateral thigh.
8. Which of the following is NOT a risk factor for developing contrast-induced nephropathy (CIN)?
   a. Type 2 diabetes mellitus
   b. Dehydration
   c. Hypertension
   d. Ventricular ejection fraction less than 50%
   e. Chronic kidney disease (CKD) (glomerular filtration rate [GFR] < 60 mL/min)
9. Nephrogenic systemic fibrosis (NSF) is:
   a. a rare genetic condition exacerbated by the use of gadolinium-based contrast medium (GBCM).
   b. immediately evident after exposure to gadolinium in 10% of exposed patients.
   c. fibrosis of the skin, subcutaneous tissue, and skeletal muscle seen in patients with chronic hypertension exposed to gadolinium contrast medium.
   d. not seen in patients with GFR greater than 60 mL/min/1.73 m^2.
   e. mainly seen in dialysis patients exposed to gadolinium contrast medium.
10. During a diuretic renal scintigraphy:

a. the diuretic is administered approximately 2 minutes after peak activity is seen in the collecting system.
b. a $T_{½}$ of greater than 14 minutes is consistent with obstruction.
c. ^{99m}Tc-DMSA is the most sensitive for obstruction and determination of GFR.
d. intestinal or gallbladder activity should never be seen with ^{99m}Tc-MAG3.
e. a $T_{½}$ of less than 10 minutes is consistent with a nonobstructed system.

11. Positron emission tomography (PET):
    a. has a higher diagnostic accuracy than CT for seminoma and nonseminoma testis cancer following chemotherapy.
    b. is sensitive and specific for detection of postchemotherapy teratoma.
    c. can be used with high positive predictive value within 2 weeks of completion of chemotherapy for bulky lymph adenopathy.
    d. has greater predictive value of primary disease in metastatic urothelial carcinoma than magnetic resonance imaging (MRI).
    e. is able to detect local or systemic recurrence of prostate cancer in 74% of patients with prostate-specific antigen recurrence.
12. What is the minimum estimated GFR for use of gadolinium-based contrast agents?
    a. Less than 30 mL/min/1.73 m^2
    b. Greater than 50 mL/min/1.73 m^2
    c. Greater than 35 mL/min/1.73 m^2
    d. Greater than 30 mL/min/1.73 m^2
    e. There are no restrictions for patients with renal insufficiency.
13. In magnetic resonance (MR) images using T2-weighted sequences, fluid appears as:
    a. dark.
    b. bright.
    c. low signal.
    d. signal void.
    e. indeterminate.

14. What lesions may have a high signal (bright) on T2-weighted MRI of the adrenal gland?
    a. Pheochromocytoma
    b. Metastasis
    c. Adrenal cortical carcinoma (ACC)
    d. None of the above
    e. All of the above
15. MR chemical shift imaging (CSI) for adrenal adenoma takes advantage of which of the following phenomena to aid in the diagnosis?
    a. Water and fat within the same voxel signals are canceled out in opposed-phase imaging.
    b. Opposed-phase imaging will exhibit a high signal (bright).
    c. Intracellular lipid content within an adenoma is low.
    d. Intravenous contrast is required.
    e. All of the above.
16. Oncocytoma typically has been characterized by a central scar. Which other renal lesion may also exhibit a central scar on T2-weighted images?
    a. Clear cell carcinoma
    b. Angiomyolipoma
    c. Chromophobe carcinoma
    d. Transitional cell carcinoma
    e. No other renal masses exhibit a central scar.
17. Which renal mass exhibits signal drop on opposed phase imaging?
    a. Papillary renal cell
    b. Chromophobe carcinoma
    c. Angiomyolipoma
    d. Clear cell carcinoma
    e. Transitional cell carcinoma
18. What signal characteristics do kidney stones exhibit on MR urography?
    a. High signal on T2-weighted images
    b. Low signal on T2-weighted images
    c. Signal void
    d. High signal on T1-weighted images

e. Low signal on T1-weighted images

19. Multiparametric imaging of the prostate consists of anatomic and functional sequences. Match the correct pair.
    a. Anatomic: Diffusion-weighted imaging
    b. Functional: T1- and T2-weighted images
    c. Anatomic: Dynamic contrast enhanced sequences
    d. Functional: Apparent diffusion coefficient maps
    e. All of the above

Answers

1. **d. Effective dose.** The distribution of energy absorption in the human body will be different based on the body part being imaged and a variety of other factors. The most important risk of radiation exposure from diagnostic imaging is the development of cancer. **The effective dose is a quantity used to denote the radiation risk (expressed in sieverts) to a population of patients from an imaging study.**
2. **e. High, 10 to 100 mSv.** The average person living in the United States is exposed to 6.2 mSv of radiation per year from ambient sources, such as radon, cosmic rays, and medical procedures, which account for 36% of the annual radiation exposure (NCRP, 2012). **The recommended occupational exposure limit to medical personnel is 50 mSv per year** (NCRP, 2012). The effective dose from a three-phase CT of the abdomen and pelvis without and with contrast may be as high as 25 to 40 mSv.
3. **e. T6.** Autonomic dysreflexia, also known as hyperreflexia, means an overactivity of the autonomic nervous system that can result in an abrupt onset of excessively high blood pressure. Persons at risk for this problem generally have spinal cord injury level above T6. Autonomic dysreflexia can develop suddenly, is potentially life threatening, and is considered a medical emergency. If not treated promptly and correctly, it may lead to seizures, stroke, and even death.

4. **b. 1 mSv.** Maintaining the maximum practical distance from an active radiation source significantly decreases exposure to medical personnel.
5. **b. Have severe renal insufficiency and take metformin the day of the study.** Patients with type 2 diabetes mellitus on metformin may have an accumulation of the drug after administering intravascular radiologic contrast medium (IRCM), resulting in biguanide lactic acidosis presenting with vomiting, diarrhea, and somnolence. This condition is fatal in approximately 50% of cases (Wiholm, 1993). [a] **Biguanide lactic acidosis is rare in patients with normal renal function. Consequently in patients with normal renal function and no known comorbidities, there is no need to discontinue metformin before IRCM use, nor is there a need to check creatinine following the imaging study.**
6. **d. The mechanism of action associated with severe idiosyncratic anaphylactoid (IA) reactions is an immunoglobulin E (IgE) antibody reaction to the contrast media.** The IA reactions are most concerning because they are potentially fatal and can occur without any predictable or predisposing factors. **Approximately 85% of IA reactions occur during or immediately after injection of IRCM and are more common in patients with a prior adverse drug reaction to contrast media; patients with asthma, diabetes, impaired renal function, or diminished cardiac function; and patients on beta-adrenergic blockers** (Spring et al., 1997).
7. **e. 0.01 mg/kg of epinephrine (1:1000 concentration), given intramuscularly in the lateral thigh.** Rapid administration of epinephrine is the treatment of choice for severe contrast reactions. Epinephrine can be administered intravenously (IV) 0.01 mg/kg body weight of 1: 10,000 dilution or 0.1 mL/kg slowly into a running IV infusion of saline and can be repeated every 5 to 15 minutes as needed. If no IV access is available, the recommended intramuscular dose of epinephrine is 0.01 mg/kg of 1:1000 dilution (or 0.01 mL/kg to a maximum of 0.15 mg of 1:1000 if body weight

is <30 kg; 0.3 mg if weight is >30 kg) injected intramuscularly in the lateral thigh.

8. **d. Ventricular ejection fraction less than 50%.** The most common patient-related risk factors for CIN are CKD (creatinine clearance <60 mL/min), diabetes mellitus, dehydration, diuretic use, advanced age, congestive heart failure, age, hypertension, low hematocrit, and ventricular ejection fraction less than 40%. **The patients at highest risk for developing CIN are those with both diabetes and preexisting renal insufficiency.**
9. **d. Not seen in patients with GFR greater than 60 mL/min/1.73 m^2.** Patients with CKD but GRF greater than 30 mL/min/1.73 m^2 are considered to be at extremely low or no risk for developing NSF if a dose of GBCM of 0.1 mmol/kg or less is used. Patients with GFR greater than 60 mL/min/1.73 m^2 do not appear to be at increased risk of developing NSF, and the current consensus is that all GBCM can be administered safely to these patients.
10. **e. a $T_{½}$ of less than 10 minutes is consistent with a nonobstructed system.** Transit time throughout the collecting system in less than 10 minutes is consistent with a normal, nonobstructed collecting system. A $T_{½}$ of 10 to 20 minutes shows mild to moderate delay and may be a mechanical obstruction. The patient's perception of pain after diuretic administration can be helpful for the treating urologist to consider when planning surgery in the patient with middle to moderate obstruction. **A $T_{½}$ of greater than 20 minutes is consistent with a high-grade obstruction.**
11. **a. Has a higher diagnostic accuracy than CT for seminoma and nonseminoma testis cancer following chemotherapy.** There are data on the use of PET/CT in testis cancer, where PET/CT was found to have a higher diagnostic accuracy than CT for staging and restaging in the assessment of a CT-visualized residual mass following chemotherapy for seminoma and nonseminomatous germ cell tumors (Hain et al., 2000; Albers et al., 1999).
12. **d. Greater than 30 mL/min/1.73 m^2. NSF occurs in patients with acute or chronic renal insufficiency with a GFR less than**

30 mL/min/1.73 m^2.

13. **b. Bright.** High signal on T2-weighted images. Fluid exhibits a low signal on T1-weighted images.
14. **e. All of the above.** Traditional teaching reported the lightbulb sign to be consistent with pheochromocytoma. However, metastasis and ACC also have a high signal on T2-weighted images. Furthermore, Varghese and colleagues reported that **35% of pheochromocytomas demonstrated low T2 signal,** contrary to conventional teaching. Therefore the conventional teaching of the "lightbulb sign" is incorrect.
15. **a. Water and fat within the same voxel signals are canceled out in opposed-phase imaging.** MR CSI is performed on T1-weighted images. Opposed-phase imaging will demonstrate a low signal (dark) if fat and water occupy the same voxel. Adrenal adenomas have high intracytoplasmic fat. CSI is performed without the use of intravenous contrast.
16. **c. Chromophobe carcinoma.** Chromophobe carcinoma exhibits a high signal on T2-weighted images.
17. **d. Clear cell carcinoma.** Microscopic intracytoplasmic lipids have been found in 59% of clear cell carcinomas, which allows it to be differentiated from other renal cell carcinoma cell types.
18. **c. Signal void.** Nephrolithiasis/calcification on MR imaging has no signal characteristics; therefore it appears as a void on imaging.
19. **d. Functional: Apparent diffusion coefficient maps.** Multiparametric MRI refers to the use of anatomic sequences (T1-weighted images, T2 triplanar [axial, sagittal, and coronal] images) and functional sequences (diffusion-weighted imaging/apparent diffusion coefficient maps, dynamic contrast-enhanced MRI, spectroscopy). The combined approach has reported negative and positive predictive values to be greater than 90% in detecting prostate cancer.

Chapter Review

1. Absorbed dose for therapy is measured in units called gray (Gy); 1 rad = 0.01 Gy, or 1 centigray (cGy) = 1 rad.
2. The amount of energy absorbed by a tissue for diagnostic purposes is referred to as the equivalent dose and is measured in sieverts (Sv). Exposure of the eyes and gonads to radiation has a more significant biologic impact than exposure of other parts of the body. The occupational safety limit is 50 mSv. Exposure time during fluoroscopy should be minimized by the use of short bursts of fluoroscopy; positioning the image intensifier as close to the patient as feasible substantially reduces scatter radiation.
3. There are four basic types of iodinated contrast media: (1) ionic monomer, (2) nonionic monomer, (3) ionic dimer, (4) nonionic dimer.
4. IA reactions are potentially fatal, are not dose dependent, and are more common in patients with a history of adverse reactions to contrast media, those with asthma or diabetes, those with impaired renal and cardiac function, and those on β-adrenergic blockers.
5. It is common to have nausea, flushing, pruritus, urticaria, headache, and occasionally emesis after administration of contrast media.
6. Patients at high risk for adverse allergic reactions should be medicated with steroids, given 12 to 24 hours before the injection of contrast media, as well as antihistamines.
7. For retrograde pyelography, it is useful to dilute contrast media by half with sterile saline, which facilitates identifying filling defects in the collecting system. There is a low risk of contrast reactions in patients in whom a retrograde or loopogram is performed.
8. Metformin does not need to be held before contrast administration in a patient with normal renal function and no comorbidities.
9. The risk of developing contrast-induced nephropathy is increased in patients with decreased renal function (GFR < 60 mL/min), diabetes mellitus, dehydration, advanced age, congestive heart failure, liver disease, and cardiac ejection fraction less than 40%.
10. TcDTPA is primarily filtered by the glomerulus. It is a good agent to assess renal function.
11. Because TcDMSA is both filtered by the glomerulus and secreted by the proximal tubule, it localizes in the renal cortex and is a good

agent for assessing cortical scarring and ectopic renal tissue.

12. TcMAG3 is cleared mainly by tubular secretion; it has a limited ability to access renal function.
13. A $T_{½}$ less than 10 minutes suggests an unobstructed system. A $T_{½}$ greater than 20 minutes is consistent with renal obstruction.
14. A positive bone scan is not specific for cancer. Moreover, the volume of cancer cannot be quantitated on bone scan. Patients with widely metastatic disease may have diffuse uptake (hyper scan) and no discrete lesions.
15. Glucose, choline, and amino acids have been used as imaging agents for PET scans.
16. ^{18}F-fluorodeoxyglucose (FDG) is used as an imaging agent in PET scanning and takes advantage of the fact that tumors have increased glycolysis and decreased dephosphorylation. This scan is useful in testicular germ cell tumors, particularly seminomas, in determining residual viable tumor following chemotherapy.
17. The Hounsfield units scale assigns a value of −1000 Hounsfield units for air. Dense bone is assigned a value of +1000 Hounsfield units, and water is assigned 0 Hounsfield units.
18. With the exception of some indinavir stones, all renal and ureteral calculi may be detected by helical CT.
19. The advantage of MRI is high-contrast resolution of soft tissue on T1-weighted images. Fluid has a low signal and appears dark on T1-weighted images; on T2-weighted images, fluid has a high signal and appears bright. Gadolinium increases the brightness of T1-weighted images. Hemorrhage within a cyst results in a high signal on T1-weighted images. MRI is the imaging modality of choice for patients with iodine contrast allergies.
20. The risk of developing NSF after gadolinium administration is increased in patients with GFRs less than 30 mL/min.
21. Adrenal adenomas have high lipid content and may be differentiated from adrenal cancers or metastatic disease by specialized CT or MRI scans.
22. Thirty-five percent of pheochromocytomas do not enhance on T2-weighted images.

23. MRI and CT are excellent imaging studies to determine the presence and extent of renal vein and vena cava tumor thrombus. Uptake of gadolinium by the thrombus on MRI differentiates tumor from bland (blood clot) thrombus.
24. Prostate MRI coupled with an assessment of dynamic contrast uptake and washout increases the diagnostic accuracy for detecting cancer.
25. MR spectroscopy for prostate cancer is based on decreased citrate levels and increased creatine and choline levels.
26. Bladder filling in patients with spinal cord injuries higher than T6 may precipitate autonomic dysreflexia.
27. Radiation exposure diminishes as the square of the distance from the radiation source.

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[a] Sources referenced can be found in *Campbell-Walsh Urology, 12th Edition,* on the Expert Consult website.

4

Urinary Tract Imaging

Basic Principles of Urologic Ultrasonography

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Keywords

sonoelastography; urinary tract imaging; urologic ultrasound

Questions

1. The maximum excursion of a wave above and below the baseline is known as its:
    a. wavelength.
    b. frequency.
    c. period.
    d. cycle.
    e. amplitude.
2. The artifact that occurs when an ultrasound wave strikes an interface at a critical angle and is refracted with limited reflection is:
    a. reverberation artifact.
    b. increased through-transmission artifact.
    c. edging artifact.
    d. comet-tail artifact.

e. aliasing artifact.

3. Which ultrasound mode allows for detection and characterization of the velocity and direction of motion?
    a. Harmonic scanning
    b. Color Doppler
    c. Power Doppler
    d. Spatial compounding
    e. Gray-scale ultrasonography
4. If the kidney is less echogenic than the liver, the kidney is described as:
    a. hyperechoic.
    b. hypoechoic.
    c. isoechoic.
    d. anechoic.
    e. echogenic.
5. The sonographic hallmark of testicular torsion is:
    a. the “blue dot” sign.
    b. epididymal edema.
    c. paratesticular fluid.
    d. increased epididymal blood flow.
    e. absence of intratesticular blood flow.
6. Ultrasound waves are examples of:
    a. radio waves.
    b. mechanical waves.
    c. electromagnetic waves.
    d. ionizing radiation.
    e. light waves.
7. The most important determinant of axial resolution is:
    a. impedance.
    b. speed of propagation.
    c. acoustic power.
    d. frequency.
    e. number of foci.
8. Increasing frequency results in a loss of:
    a. absorption.
    b. axial resolution.

c. lateral resolution.
d. depth of penetration.
e. mechanical index.

9. When sound waves encounter the interface between two tissues with large differences in impedance, the waves are:
   a. increased in frequency.
   b. decreased in frequency.
   c. reflected.
   d. refracted.
   e. reverberated.
10. When a tissue appears darker than the surrounding tissue on ultrasound it is said to be relatively:
   a. hypoechoic.
   b. hyperechoic.
   c. hypodense.
   d. isoechoic
   e. anechoic.
11. The focal zone represents the area of best:
   a. lateral resolution.
   b. axial resolution.
   c. echogenicity.
   d. blood flow.
   e. tissue penetration.
12. Increasing the gain has the effect of:
   a. increasing amplitude of the sound waves.
   b. increasing acoustic power.
   c. increasing thermal index.
   d. increasing mechanical index.
   e. increasing transducer sensitivity.
13. One way to improve the visualization of deep structures is to:
   a. increase the frequency.
   b. decrease the frequency.
   c. increase the wave velocity.
   d. decrease the gain.
   e. employ Doppler flow.

14. The best frequency for performing external renal ultrasound in most adults is:
    a. 3.5 to 5 MHz.
    b. 6 to 7 MHz.
    c. 7.5 to 8.4 MHz.
    d. 8.5 to 9.9 MHz.
    e. None of the above.
15. A simple cyst of the kidney would not display which of the following characteristics?
    a. Bright back wall
    b. Increased through transmission
    c. Anechoic interior
    d. Edging artifact
    e. Hyperechoic internal nodule
16. Which of the following is correct?
    a. Measuring bladder volume requires three-dimensional scanning.
    b. A nearly empty bladder is desirable for bladder scanning.
    c. A curved array transducer is preferred for bladder ultrasound in most patients.
    d. Ureteroceles are usually poorly visualized because the membrane is thin.
    e. Bladder ultrasound is a sensitive screening exam for suspected bladder tumors.
17. Which of the following are evaluable by transabdominal bladder ultrasound?
    a. Urine volume
    b. Bladder wall characteristics
    c. Stones or diverticulum
    d. Dilated ureters
    e. All of the above
18. Scrotal ultrasound for the evaluation of possible testicular torsion may include all of the following but must include:
    a. B-mode ultrasound.
    b. multiple scrotal views.
    c. Doppler flow studies.

d. simultaneous bilateral views.
e. harmonic scanning.

19. The most important limitation of ultrasound in attempting to characterize complex renal cysts as benign or malignant is:
    a. refraction.
    b. inability to evaluate enhancement.
    c. lack of axial resolution.
    d. increased through-transmission with artifact.
    e. reverberation artifact.
20. A complete transrectal ultrasound of the prostate should include an evaluation of:
    a. rectal wall.
    b. seminal vesicles and ejaculatory ducts.
    c. bladder.
    d. prostate.
    e. all of the above.
21. Which of the following would not typically be visible in a sagittal midline prostate ultrasound?
    a. Middle lobe of the prostate
    b. Central zone
    c. Ejaculatory duct
    d. Tip of right seminal vesicle
    e. Apex of the prostate
22. Characteristics of prostate cancer as demonstrated on transrectal ultrasound of the prostate may include:
    a. hypoechogenicity.
    b. hyperechogenicity.
    c. prostate asymmetry.
    d. increased vascularity.
    e. all of the above.
23. The single most important determinant of patient safety in ultrasound utilization is:
    a. multifrequency probes.
    b. good documentation.
    c. an informed operator.
    d. periodic equipment inspection.

e. Doppler capability.

24. The disinfection level protocol required for transrectal ultrasound probes following needle biopsy is:
    a. low.
    b. moderate.
    c. high.
    d. critical.
    e. none of the above.
25. For patient safety it is preferable to maximize and limit:
    a. acoustic power, exam duration.
    b. acoustic power, gain.
    c. gain, acoustic power.
    d. gain, use of cine function.
    e. images, description.

Answers

1. **e. Amplitude.** In ultrasound physics it is crucial to understand the concept of amplitude. The amplitude of an ultrasound wave represents its relative energy state, and it is the amplitude of the returning sound wave that determines the pixel brightness to be displayed on a monitor during real-time gray-scale imaging.
2. **c. Edging artifact.** Echo reflection is the primary mechanism whereby sound waves are returned to a transducer. It is important to understand how the angle of insonation affects the reflection and refraction of sound waves. There is a critical angle at which waves will travel along an interface rather than being returned to the probe. When this angle is encountered, it provides a dark or hypoechoic “shadow” called an “edging artifact.” A reverberation artifact is one caused by multiple transits of a sound wave between the transducer and the reflecting object. Increased through-transmission artifact is caused by decreased attenuation of sound waves as they travel through a fluid-filled structure. Comet-tail artifact is seen as the result of the interaction between sound waves

and fluid and gas filled structures such as the bowel. Aliasing artifact is seen with Doppler ultrasonography.

3. **b. Color Doppler.** Doppler ultrasonography is important for evaluating motion and flow. The critical difference between color Doppler and power Doppler is that color Doppler is able to evaluate both flow velocity and direction. Power Doppler evaluates integrated amplitude of the returning sound waves. Although gray-scale ultrasonography does permit the evaluation of motion, it does not permit the characterization of velocity or direction. Harmonic scanning and spatial compounding are modes that allow the selective evaluation or combination of reflected frequencies in ways that improve image quality.
4. **b. Hypoechoic.** In describing ultrasound images, it is important to use correct terminology. Descriptive terms involving echogenicity are relative terms. A hyperechoic or hypoechoic structure is being described in relation to the echogenicity of a reference standard. In most cases the reference standard is the liver. In the adult the normal kidney is hypoechoic relative to the normal liver in approximately 75% of patients.
5. **e. Absence of intratesticular blood flow.** The absence of intratesticular blood flow is the classic sonographic finding in testicular torsion. However, there are many documented cases of some preserved intratesticular blood flow even in cases with significant torsion. Therefore testicular torsion remains a clinical diagnosis. Epididymal edema, paratesticular fluid, and increased epididymal blood flow may be seen with testicular torsion but may also be seen with other clinical conditions. The blue dot sign is a classic physical finding in torsion of the appendix testis.
6. **b. Mechanical waves:** Mechanical waves are represented graphically as a sine wave alternating between a positive and negative direction from the baseline. Sound waves as they propagate through human tissue produce areas of returning compression and rarefaction.
7. **d. Frequency:** Axial resolution is directly dependent on the frequency of sound waves. The higher the sound wave's frequency is, the better the axial resolution.

8. **d. Depth of penetration:** The optimal ultrasound image requires tradeoffs between resolution and depth of penetration. High-frequency transducers of 6 to 10 MHz may be used to image structures near the surface of the body (e.g., testis, pediatric kidney) with excellent resolution. However, deeper structures (e.g., right kidney, bladder) require lower frequencies of 3.5 to 5 MHz to penetrate. Such images will have poorer axial resolution.
9. **c. Reflected:** The shape and size of the object and the angle at which the advancing wave strikes the object are critical determinants of the amount of energy reflected. The amount of energy reflected from an interface is also influenced by the **impedance** of the two tissues at the interface. Impedance is a property that is influenced by tissue stiffness and density. It is the difference in impedance that allows an appreciation of interfaces between different types of tissue.
10. **a. Hypoechoic:** The liver is usually used as a benchmark for echogenicity. A hypoechoic area is described as “darker” on B-mode ultrasound.
11. **a. Lateral resolution:** Lateral resolution refers to the ability to identify separately objects that are equidistant from the transducer. Lateral resolution is a function of the focused width of the ultrasound beam and is a characteristic of the transducer. The location of the narrowest beam width can be adjusted by the user. The more focused the beam is, the better the lateral resolution at that location. Thus image quality can be enhanced by locating the narrowest beam width (focus or focal zone) at the depth of the object or tissue of interest
12. **e. Increasing transducer sensitivity:** The gain control on the console of the ultrasound machine permits the user to increase or decrease the sensitivity of the transducer to reflected sound waves.
13. **b. Decrease the frequency:** The optimal ultrasound image requires trade-offs between resolution and depth of penetration. High-frequency transducers of 6 to 10 MHz may be used to image structures near the surface of the body (e.g., testis, pediatric kidney), with excellent resolution. However, deeper structures

(e.g., right kidney, bladder) require lower frequencies of 3.5 to 5 MHz to penetrate. Such images will have poorer axial resolution.

14. **a. 3.5 to 5 MHz:** Deeper structures (e.g., right kidney, bladder) require lower frequencies of 3.5 to 5 MHz to penetrate.
15. **e. Hyperechoic internal nodule:** A simple cyst is an example of a structure that is well circumscribed, with an anechoic interior and through transmission.
16. **c. A curved array transducer is preferred for bladder ultrasound in most patients:** A curved array transducer is of lower frequency (3.5–6 MHz) and provides greater depth of penetration but with less axial resolution. It is most often the transducer of choice for imaging the kidney and urinary bladder.
17. **e. All of the above:** Urine volume, bladder wall characteristics, the presence of calculi or diverticulum, and the presence of dilated ureters just outside the bladder are all evaluable by transabdominal bladder ultrasound.
18. **c. Doppler flow studies:** Caution should be used when interpreting Doppler flow studies in the evaluation of suspected testicular torsion. The hallmark of testicular torsion is the absence of **intratesticular** blood flow. Paratesticular flow in epididymal collaterals may appear within hours of torsion. Comparison with the contralateral testis should be performed to ensure that the technical attributes of the study are adequate to demonstrate intratesticular blood flow.
19. **b. Inability to evaluate enhancement:** Unlike computed tomography (CT), currently ultrasound cannot evaluate enhancement. Once it is approved by the US Food and Drug Administration (FDA) for this purpose, contrast-enhanced ultrasound will allow the detection of enhancement of renal masses by ultrasound.
20. **e. All of the above:** When evaluating the prostate, surrounding structures need to be assessed. Rectal lesions (including cancer), dilated seminal vesicles, and/or ejaculatory ducts as well as bladder pathology should all be evaluated for a complete examination.
21. **d. Tip of right seminal vesicle:** In a midline sagittal view the tips of the seminal vesicles are not normally visualized on ultrasound.

An axial projection needs to be obtained to be able to measure the length of each seminal vesicle.

22. **e. All of the above:** Although excellent resolution and tissue characteristics is possible with transrectal ultrasound a diagnosis of prostate cancer is not often able to be made with ultrasound alone.
23. **c. An informed operator:** The ALARA (as low as reasonably achievable) principle is intended to limit the total energy imparted to the patient during an examination. This can be accomplished by (1) keeping power outputs low, (2) using appropriate scanning modes, (3) limiting examination times, (4) adjusting focus and frequency, and (5) using the cine function during documentation. All of these are dependent upon an informed sonographer.
24. **d. Critical:** Any time body fluids or tissues come in contact with an ultrasound transducer, critical or high-level disinfection protocols must be strictly adhered to.
25. **c. Gain, acoustic power:** Unlike gain, which refers to amplification of the acoustic signal returning to the transducer, acoustic power is the amount of acoustic energy applied to the tissue. The biologic effects of ultrasound in terms of power are in the milliwatt range. High levels generate heat and cavitation, which might result in tissue damage.

Chapter Review

1. One cycle per second is known as 1 hertz (Hz). High-frequency ultrasonic transducers of 6 to 10 MHz are used to image structures near the surface. Deeper structures require lower frequencies of 3.5 to 5 MHz. Axial resolution improves with increasing frequency, and depth of penetration decreases with increasing frequency.
2. Resistive index is the peak velocity minus the end-diastolic velocity over the peek systolic velocity. This is measured using the color flow Doppler with spectral display and is used to characterize renal artery stenosis, ureteral obstruction, and penile arterial insufficiency.
3. By convention, the liver is used as a benchmark for echogenicity. By convention, the cephalad aspect of the structure is to the left of the

image.

4. Ultrasonography may produce injury due to mechanical effects caused by cavitation or by heat generation.
5. With the exception of some indinavir stones, all renal and ureteral calculi may be detected by helical CT.
6. The advantage of magnetic resonance imaging (MRI) is high-contrast resolution of soft tissue on T1-weighted images. Fluid has a low signal and appears dark on T1-weighted images; on T2-weighted images, fluid has a high signal and appears bright. Gadolinium increases the brightness of T1-weighted images. Hemorrhage within a cyst results in a high signal on T1-weighted images.